



Figure 3: dbShards

to bring the benefits of the relational model to the distributed architecture, and improve the performance of relational databases to an extent that the scalability is no longer an issue.” Figure 3 is an architectural example of one of the NewSQL solutions (dbShards).

Technical characteristics of NewSQL solutions

1. SQL as the primary mechanism for application interaction.
2. ACID support for transactions.
3. A non-locking concurrency control mechanism so real-time reads will not conflict with writes, and thus cause them to stall.
4. An architecture providing much higher per-node performance than available from traditional RDBMS solutions.
5. A scale-out, shared-nothing architecture, capable of running on a large number of nodes without suffering bottlenecks.

The expectation is that NewSQL systems are about 50 times faster than traditional OLTP RDBMS.

Similar to NoSQL, there are many categories of NewSQL solutions. Now we can look at these categorisations and their characteristics.

NewSQL categorisation

Categorisation is based on the different approaches adopted by vendors to preserve the SQL interface, and address the scalability and performance concerns of traditional OLTP solutions.

1. **New databases:** These NewSQL systems are newly designed from scratch to achieve scalability and performance. Of course, some (hopefully minor) changes to the code will be required and data migration is still needed. One of the key considerations in improving the performance is making non-disk (memory) or new kinds of disks (flash/SSD) the primary data store. Solutions can be software-only (VoltDB, NuoDB and Drizzle) or supported as an appliance (Clustrix, Translatice). Examples of offerings are Clustrix, NuoDB and Translatice (commercial); and VoltDB, Drizzle, etc (open source).

2. **New MySQL storage engines:** MySQL is part of the LAMP stack and is used extensively in OLTP. To overcome MySQL's scalability problems, a set of storage engines are developed, which include Xeround, Akiban, MySQL NDB cluster, GenieDB, Tokutek, etc. The good part is the usage of the MySQL interface, but the downside is data migration from other databases (including old MySQL) is not supported. Examples of offerings are Xeround, GenieDB and Tokutek (commercial); and Akiban, MySQL NDB Cluster and others in open source.
3. **Transparent clustering:** These solutions retain the OLTP databases in their original format, but provide a pluggable feature to cluster transparently, to ensure scalability. Another approach is to provide transparent sharding to improve scalability. Schooner MySQL, Continuent Tungsten and ScalArc follow the former approach, whereas ScaleBase and dbShards follow the latter approach. Both approaches allow reuse of existing skillsets and ecosystem, and avoid the need to rewrite code or perform any data migration. Examples of offerings are ScalArc, Schooner MySQL, dbShards and ScaleBase (commercial); and Continuent Tungsten (open source).

Summing up

Given the continuing trend of data growth in OLTP systems, a new generation of solutions is required to cater to them. DBMSs have to be re-architected from scratch to meet this demand. However, unlike NoSQL, these DBMSs have to cater for applications already written for an earlier generation of RDBMS. Hence, drastic interface changes like throwing out SQL or extensive changes to data schema is out of the question. A new generation of information management systems, termed NewSQL systems, caters to this trend and these constraints. NewSQL is apt for businesses that are planning to: (1) migrate existing applications to adapt to new trends of data growth, (2) develop new applications on highly scalable OLTP systems, and (3) rely on existing knowledge of OLTP usage.

We look forward to your comments on this introductory article, and we will also plan for a series of expert in-depth articles on the NewSQL database technology. 

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